Research Cotton seeds and seed oil cause infertility in men, and have been tested as a male contraceptive in China. However, in addition to lowering sperm count, cotton seed oil causes the degeneration of sperm-producing cells.
Related Species The American species *G. hirsutum* was used extensively as a medicinal herb by the Maya and Aztecs, and was also cultivated for its fiber. Columbus carried samples of this species back to Europe from his first voyage. Native American people used the bark to ease the pain of childbirth, and by the 19th century it was used as an inducer of menstruation and abortion.
Cautions Cotton root bark and seed oil are potentially toxic and should only be used under professional supervision. Do not use during pregnancy.

Grindelia camporum syn. *G. robusta* var. *rigida* (Asteraceae)

Gumplant

Description Perennial herb growing to 3 ft (1 m). Has triangular leaves and yellow-orange daisy-type flowers.
Habitat & Cultivation Native to the southwestern U.S. and Mexico, gumplant grows in arid and saline soil. It is harvested in late summer when in flower.
Parts Used Leaves, flowering tops.
Constituents Gumplant contains diterpenes (including grindelic acid), resins, and flavonoids.
History & Folklore Gumplant was used by Native Americans to treat bronchial problems and skin afflictions. Gumplant was officially recognized in the *Pharmacopoeia of the United States* from 1882 to 1926.
Medicinal Actions & Uses Gumplant is a valuable remedy for bronchial asthma, and for states where phlegm in the airways impedes respiration. Both antispasmodic and expectorant, gumplant helps to relax the muscles of the smaller bronchial passages and to clear congested mucus. Additionally, it is thought to desensitize the nerve endings in the bronchial tree and to slow the heart rate, both leading to easier breathing. Gumplant is also taken for bronchitis and emphysema, and to clear mucus buildup in the throat and nose. It has been employed in the treatment of whooping cough, hay fever, and cystitis, and externally to help speed the healing of skin irritation and burns.
Related Species *G. squarrosa*, a North American species used interchangeably with *G. camporum*, was taken by Native Americans to treat respiratory problems such as colds, coughs, and tuberculosis.
Cautions Toxic in excessive doses. Only take under professional supervision. Do not take if suffering from kidney or heart problems.

Guaiacum officinale (Zygophyllaceae)

Lignum Vitae,

Guayacan (Spanish)

Description Evergreen tree growing to 10 m (33 ft). Has compound oval leaves, small, deep blue star-shaped flowers, and heart-shaped seed capsules.
Habitat & Cultivation Lignum vitae is native to South America and the Caribbean islands. It grows in tropical rainforests. The tree is felled for its timber, and resin is extracted from the heartwood.
Parts Used Wood, resin.
Constituents Lignum vitae contains lignans (furoguaiacidin, guaiacin, and others), triterpene saponins, 18–25% resin, and volatile oil.

***Lignum vitae** was once in high demand in Europe as a purported cure for syphilis.*

History & Folklore In 1519, Ulrich von Hutten, a German satirist, was said to have cured himself of syphilis after a 40-day regimen involving fasting, profuse sweating, and drinking decoctions of lignum vitae. Furthermore, in 1526, Oviedo, one of the earliest chroniclers of American natural history, wrote that "Caribbean Indians cure themselves very easily" of venereal disease with this plant. For some years, lignum vitae was in great demand in Europe but it slowly fell into disrepute, its use as a cure for syphilis being seen as a long-lasting hoax. However, it is possible that the herb might have some effect if combined with an intensive naturopathic regimen.
Medicinal Actions & Uses Used in Europe, especially in Britain, as a remedy for arthritic and rheumatic conditions, lignum vitae has anti-inflammatory properties that help to reduce joint pain and swelling. It is also diuretic, laxative, and sweat-inducing and speeds the elimination of toxins, which makes it valuable for treating gout. Tincture of lignum vitae is commonly used as a friction rub on rheumatic areas. Absorbent cotton moistened with the resin may be applied to aching teeth. A decoction of the woodchips acts as a local anesthetic, and is used to treat rheumatic joints and herpes blisters.
Related Species *G. sanctum*, which grows in Central America and parts of Florida, and *G. coulteri*, native to Mexico, are used in the same manner as lignum vitae.
Caution Lignum vitae is subject to legal restrictions in some countries and is endangered.

Guarea rusbyi syn. *G. guidonia* (Meliaceae)

Cocillana,

Guapi Bark

Description Evergreen tree growing to 150 ft (45 m) with pale grey bark, compound lance-shaped leaves, and green-white flowers.
Habitat & Cultivation Cocillana is native to the eastern Andes. The bark is gathered throughout the year.
Part Used Bark.
Constituents Cocillana contains anthraquinones, proanthocyanids, and a volatile oil.
History & Folklore Cocillana has been used as an emetic in traditional South American and Caribbean medicine, probably for many centuries. The plant was first introduced to Western medicine by H. H. Rusby, who collected samples in Bolivia in 1886.
Medicinal Actions & Uses Cocillana is used in cough mixtures, being an even more powerful expectorant than ipecac (*Cephaelis ipecacuanha*, p. 186). Cocillana is taken as a treatment for coughs, excessive mucus production in the throat and chest, and bronchitis. At a high dosage, the plant induces vomiting.
Related Species A gum resin derived from the Caribbean *G. guara* is used as a clotting agent, and a decoction of the leaves is taken as a treatment for internal bleeding.
Caution Use cocillana only under professional supervision.

Gymnema sylvestre (Asclepiadaceae)

Gymnema,

Gurmar (Hindi)

Description Large, evergreen, twining plant, climbing up through forest trees, sometimes to a considerable height. Has dull green leaves about 2 in (5 cm) long, and umbels of small yellow flowers.